Q1. Explain what Tableau is and why it is preferred over traditional spreadsheet tools for data analysis.Also mention any two industries where Tableau is commonly used and explain one use case.

Ans: Compared to tools like Microsoft Excel, Tableau offers several advantages:

1. Better data visualization

Tableau creates interactive and visually appealing charts, dashboards, and graphs, while spreadsheets are mostly static.

2. Handles large datasets

Excel struggles with very large data, but Tableau can process millions of rows efficiently.

3. Easy drag-and-drop interface

No need for complex formulas—users can build charts simply by dragging fields.

4. Real-time data connection

Tableau can connect directly to databases and update dashboards in real time, unlike spreadsheets that often require manual updates.

5. Advanced analytics

It supports features like forecasting, trend analysis, and data blending without heavy coding.

Industries where Tableau is commonly used

1. Healthcare

- Used to track patient data, hospital performance, and treatment outcomes

2. Retail

- Used to analyze sales, customer behavior, and inventory trends

Example use case (Retail industry)

A retail company uses Tableau to:

- Track **daily sales across regions**
- Identify **top-selling products**
- Analyze **customer purchase patterns**

This helps managers make decisions like:

- Which products to restock
- Which region needs marketing focus

Q2. After connecting to the Sample – Superstore dataset:

a) List the main types of fields available (examples: time-based, geographical, numerical).

b) Identify any four Dimensions and any four Measures from the dataset.

Hint: Look at the Data Pane on the left side of Tableau and observe blue (Dimensions) and green (Measures) fields.

Ans: **a) Main types of fields available**

1. Time-based fields

- Used for analyzing trends over time
- Example: Order Date, Ship Date

2. Geographical fields

- Used for maps and location-based analysis
- Example: Country, State, City, Postal Code

3. Numerical fields

- Contain quantitative values used for calculations
- Example: Sales, Profit, Quantity

4. Categorical fields

- Used to group and segment data
- Example: Category, Sub-Category, Segment

b) Four Dimensions and Four Measures

Dimensions (blue fields)

These are categorical/descriptive:

1. Order ID
2. Customer Name
3. Region
4. Category

Measures (green fields)

These are numerical/aggregated:

1. Sales

2. Profit
3. Quantity
4. Discount

Summary

- **Dimensions** → define *how you break the data*
- **Measures** → define *what you calculate*

This distinction is what allows Tableau to quickly build visualizations just by dragging fields into the view.

Q3. Create a Bar Chart showing Total Sales by Category.

Explain: Which field goes to Rows

Which field goes to Columns

Which aggregation is applied to Sales

Hint: Drag Category and Sales into the view and observe Tableau's default behavior.

Ans:

1. Which field goes to Rows?

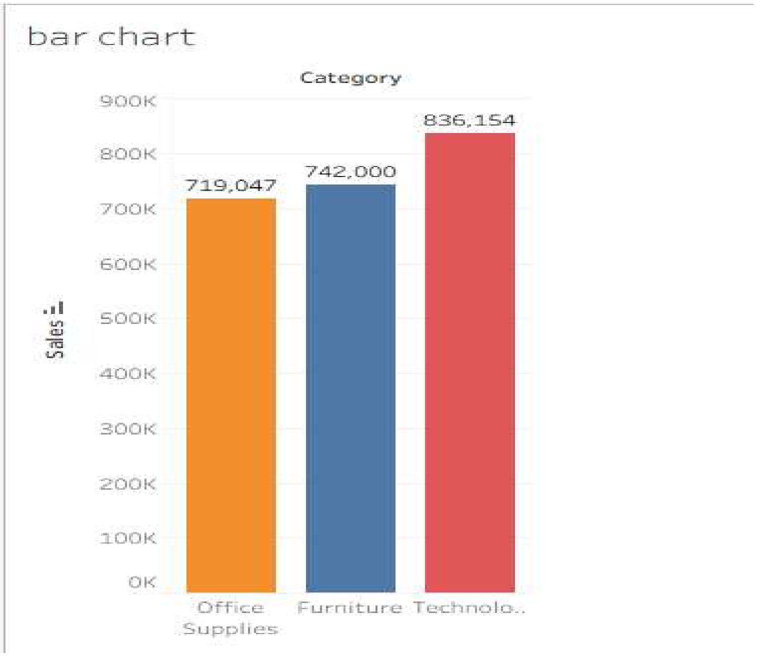
- **Category** goes to the **Rows shelf**
- This creates separate rows for each category (like Furniture, Office Supplies, Technology)

2. Which field goes to Columns?

- **Sales** goes to the **Columns shelf**
- This creates the horizontal axis showing sales values

3. Which aggregation is applied to Sales?

- Tableau applies **SUM (Sales)** by default
- That means it calculates **total sales for each category.**



Q4. What insight can be gained by comparing Sales and Profit together?

Then explain how you would create a chart to analyze Profit by Sub-Category.

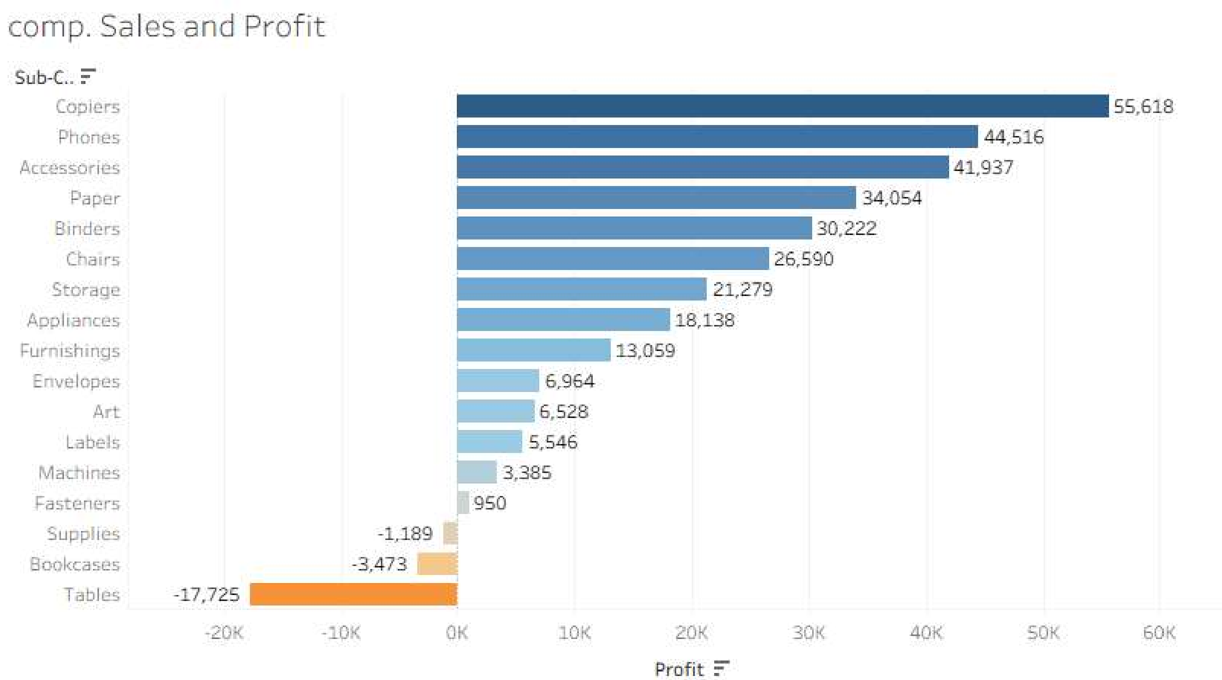
Hint: Use Sub-Category as a Dimension and Profit as a Measure.

Ans: **Insight from comparing Sales and Profit**

Looking at both together reveals:

- **High Sales but Low/Negative Profit**
  Indicates products that sell well but are not profitable (maybe due to high discounts or costs)
- **Low Sales but High Profit**
  Niche products that generate strong margins
- **Balanced High Sales & High Profit**
  Ideal products/categories to focus on

Example insight:
A category may show strong sales numbers but poor profit, meaning the business is **earning revenue but not making money efficiently**.



Q5. Create a Line Chart to show Sales Trend over Order Date.

Which date level would you choose (Year / Quarter / Month)?

Why are line charts suitable for time-based analysis?

Hint: Drag Order Date and change its date hierarchy from Year to Month if needed.

Ans: **Level selected Month: Why Month?**

Year → too high-level, hides trends

Quarter → moderate detail

Month → best balance of detail and clarity

Month helps to see:

Seasonal patterns

Growth/decline trends

Short-term fluctuations

Why line charts are suitable for time-based analysis

- Show trends clearly over time
- Display continuous data (time flows naturally)
- Help identify:

- Upward or downward movement
- Seasonal patterns
- Sudden spikes or drops

The connected line makes it easy to follow changes from one time point to the next.

Q6. Apply a Basic Filter to display data only for:

Region = “West”

Category = “Technology”

Explain the steps and mention which shelf is used.

Hint: Drag the required fields to the Filters shelf and select appropriate values.

Ans: **Steps to apply the filter**

1. Drag fields to Filters shelf

- Drag **Region** → **Filters shelf**
- In the selection window, choose **West**
- Drag **Category** → **Filters shelf**
- Select **Technology**

2. Confirm the selections

- Click **OK** after selecting values
- Tableau will immediately update the view to show only filtered data

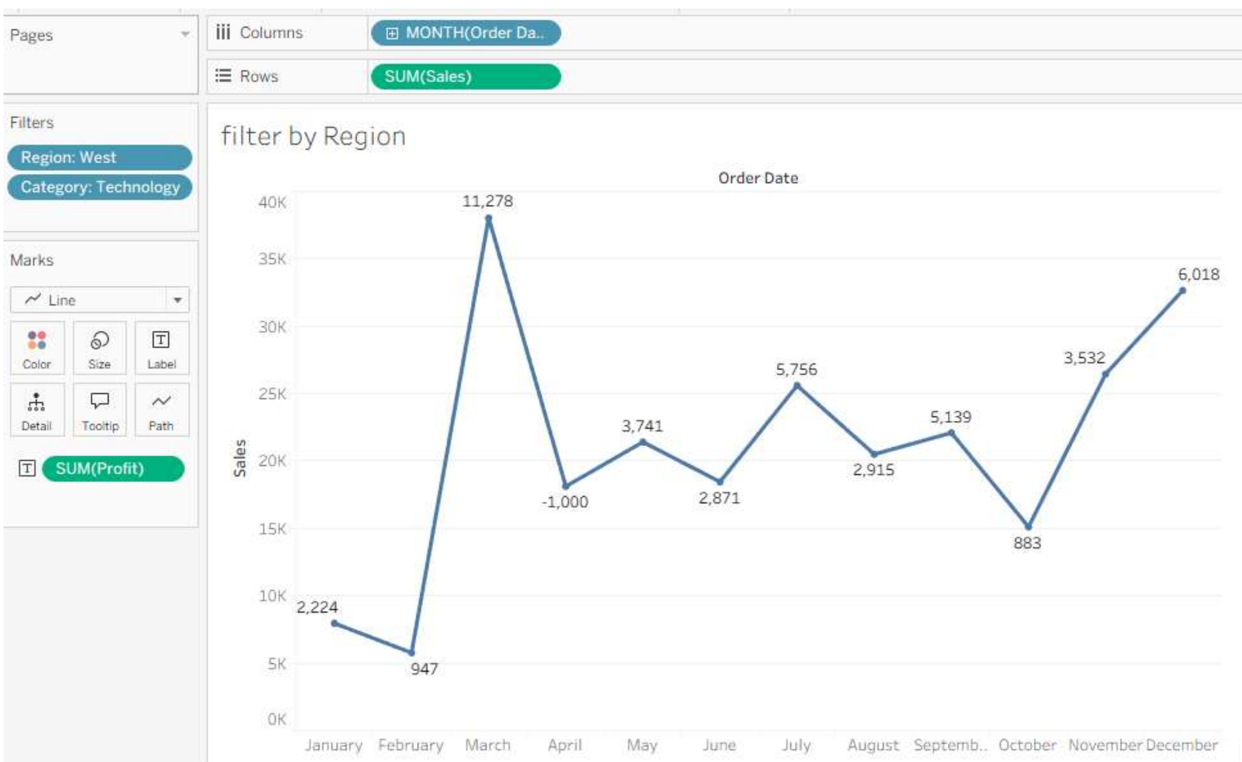
Which shelf is used?

The **Filters shelf** is used to apply these conditions.

What happens after filtering?

- Only records where:
    - Region = West
    - Category = Technology
      will remain visible in the visualization.

All other data is excluded from the view.



Q7. Explain the business importance of geographical analysis.

Then describe how you would create a Map showing Sales by State using Sample – Superstore.

Hint: Use State and Sales. Tableau automatically detects geographic fields.

Ans: **Business importance of geographical analysis**

Geographical analysis is important because it helps organizations:

1. Identify high and low-performing regions

- Shows where sales are strong or weak
- Helps focus efforts on underperforming areas

2. Improve decision-making

- Businesses can allocate resources (marketing, stock, staff) based on location performance

3. Optimize supply chain and distribution

- Helps decide where to store inventory or open new warehouses

4. Understand customer behavior by region

- Different regions may have different buying patterns or preferences

Example: A company may discover that "West" region has high sales but "South" needs improvement.

Creating a Map: Sales by State (Sample – Superstore)

Using the **Sample – Superstore** dataset:

Step 1: Use geographic field

- Drag **State** → view area

Tableau automatically recognizes it as a **geographic role**

Step 2: Add Sales measure

- Drag **Sales** → Marks card → Color (or Size)

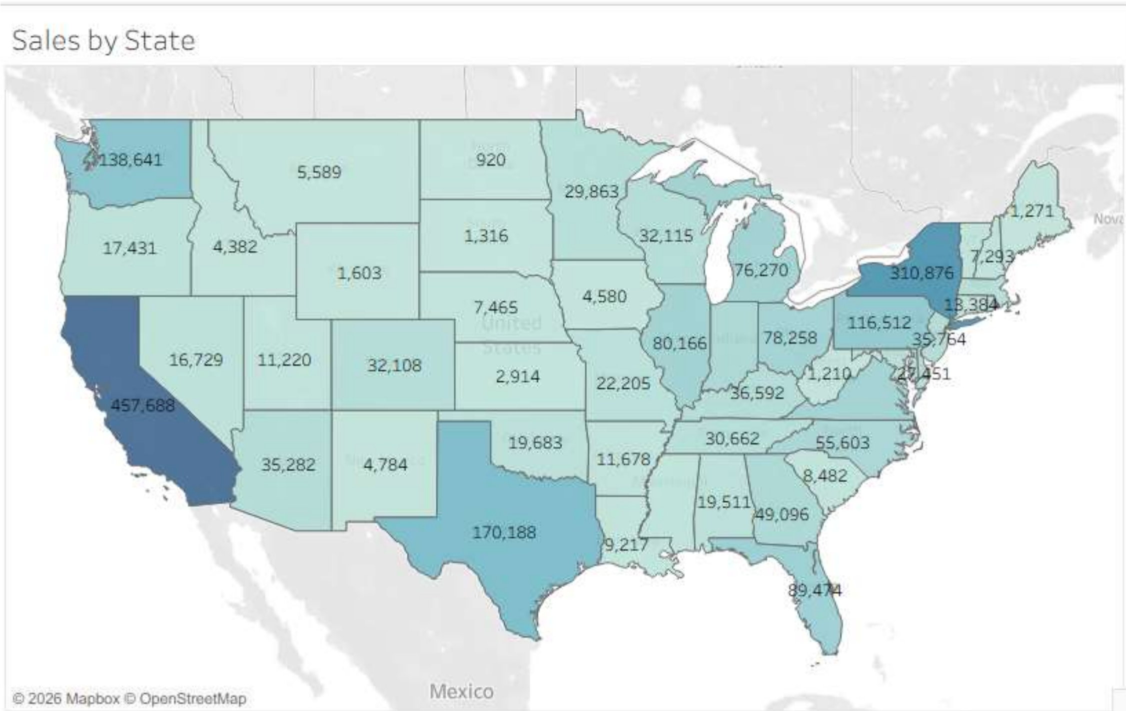
Step 3: Generate the map

- Tableau automatically creates a **filled map or symbol map**

What the map shows

- Each state is displayed on the map
- Color intensity represents **total Sales**

- Darker color = higher sales
- Lighter color = lower sales



Q8. What is the difference between a Worksheet and a Dashboard in Tableau?

Why is a dashboard more useful for decision-makers?

Ans: **1. Worksheet**

A **worksheet** is a single view where you build one visualization.

- Contains **one chart at a time** (bar chart, line chart, map, etc.)
- Used for **detailed analysis**
- Focuses on a specific question (e.g., “Sales by Region”)

Example:

- A bar chart showing Sales by Category

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2. Dashboard

A **dashboard** is a collection of multiple worksheets combined into one screen.

- Can contain **multiple charts, maps, and filters**

- Provides a **complete view of data**
- Allows interaction between visuals

Example:

- A dashboard showing:
    - Sales by Region (map)
    - Profit by Category (bar chart)
    - Sales trend over time (line chart)

Why dashboards are more useful for decision-makers

1. Complete view of business data

Decision-makers can see **all key metrics in one place** instead of switching between sheets.

2. Faster decision-making

Multiple insights (sales, profit, trends) are visible together, saving time.

3. Interactive analysis

Dashboards allow:

- Filters
- Highlight actions
- Drill-down analysis

4. Better storytelling

Combining visuals helps explain **business performance clearly and visually**.

5. Real-time monitoring

Dashboards can connect to live data for **up-to-date insights**.

Final summary

- **Worksheet = one visualization for analysis**
- **Dashboard = collection of visualizations for overview and decision-making**
- Dashboards are more powerful because they provide a **complete, interactive business picture**

Q9. What is Tableau Prep, and why is it important in real-world data projects?

Then explain how you would:

Remove null values

Remove duplicate records

Change incorrect data types (Answer in steps, no tool access required.)

Ans: Tableau Prep is a **data preparation tool** used to clean, reshape, and organize raw data before it is used for analysis in Tableau.

It is important because real-world data is often:

- Incomplete (missing values)
- Messy (duplicates, errors)
- Unstructured (wrong formats or types)

Tableau Prep helps make data **clean, consistent, and analysis-ready**.

Why Tableau Prep is important in real-world projects

1. Improves data quality

Ensures analysis is based on **accurate and clean data**

2. Saves time

Automates cleaning tasks instead of manual spreadsheet work

3. Handles large datasets

Works well with complex, multi-source data

4. Prepares data for analysis

Feeds clean data into Tableau dashboards for better insights

How to clean data in Tableau Prep (conceptual steps)

1. Remove null values

Null values = missing data

Steps:

1. Load dataset into Tableau Prep
2. Click on the column with null values
3. Use **Filter step**

4. Select option:
    - Exclude NULL values OR
    - Use “Clean Step” → remove null rows
5. Apply changes

Result: Dataset will no longer contain missing entries in that field

2. Remove duplicate records

Duplicates = repeated rows

Steps:

1. Add a **Clean Step**
2. Select the fields that define uniqueness (e.g., Order ID, Customer ID)
3. Click **Remove Duplicates option**
4. Tableau Prep keeps only one record per duplicate group

Result: Each record becomes unique

3. Fix incorrect data types

Example: Numbers stored as text, dates in wrong format

Steps:

1. Click on the field in Tableau Prep
2. Check the data type icon (Abc, #, calendar)
3. Click dropdown next to field
4. Select correct type:
    - String → Number
    - String → Date
    - Number → String (if needed)
5. Validate changes

Result: Data becomes consistent and usable for calculations

Q10. A retail manager wants answers to the following:

Which category generates the highest profit?

Which region performs poorly?

How does sales change over time?

a) Mention which charts you would use

b) Identify one key insight that management could derive

c) Explain how this helps in business decision-making

Ans: **a) Which charts you would use**

1. Which category generates the highest profit?

Bar Chart

- Dimension: Category
- Measure: Profit (SUM)

Purpose: Easy comparison between categories

2. Which region performs poorly?

Map (Filled Map)

- Dimension: Region (or State)
- Measure: Profit or Sales

Purpose: Identify weak-performing locations visually

3. How does sales change over time?

Line Chart

- Dimension: Order Date (Month/Year)
- Measure: Sales (SUM)

Purpose: Shows trends over time

b) One key insight management could derive

Example insight:

- "Technology category generates the highest profit, but some regions like the South are underperforming, and sales show seasonal fluctuations over time."

This helps identify:

- Best-performing products
- Weak geographic areas
- Seasonal demand patterns

c) How this helps in business decision-making

1. Better resource allocation

- Invest more in high-profit categories
- Improve marketing in weak regions

2. Strategic planning

- Focus on improving low-performing regions
- Expand successful product categories

3. Trend-based forecasting

- Sales trend helps predict future demand
- Supports inventory planning

4. Faster data-driven decisions

- Visual insights are quicker than reading raw data tables

Final summary

- Bar chart → Profit by Category
- Map → Performance by Region
- Line chart → Sales over Time
- Insight → Identifies strengths, weaknesses, and trends
- Benefit → Supports smarter, data-driven business decisions